

Bastion-Shield.AI (BS.AI)

Makami.AI — Standalone Multi-Agent Health Wrapper Specification

- SPEC v0.4
- STANDALONE WRAPPER
- FAIL-CLOSED ARCHITECTURE
- NON-SLM

Executive Summary: Bastion-Shield.AI (BS.AI) is an independent, deterministic multi-agent health wrapper engineered for high-assurance environments. It operates strictly as a thermodynamic and policy validation boundary—it is neither a Small Language Model (SLM) nor a custom cryptographic cipher. Host systems pass a standardized *HealthSnapshot*, and the wrapper enforces structural integrity, hop caps, four-cache telemetry, the Hideo paranoia filter, and Minuteman dissipate-and-stand-down protocols without ever modifying or generating computational state.

1. Core Pipeline Architecture

Every message passing through the BS.AI wrapper traverses a strict 8-phase validation pipeline mapped from traditional martial arts principles to ensure fail-closed security:

Shin → Rei → Kumi-kata → Kuzushi → Tsukuri → Kake → Ukemi → Zanshin

Phase	Operational Function & Security Enforcement
Shin (Mind)	Core initialization and agent registry lookup. Establishes baseline execution context.
Rei (Etiquette)	Actor verification. Rejects unknown actors and enforces quarantine constraints for non-live agents.
Kumi-kata (Gripping)	Message typing validation. Refuses untyped or unlisted message structures, triggering immediate Minuteman dissipation (<i>RT-4</i>).
Kuzushi (Unbalancing)	Hop-count enforcement and Hideo filter evaluation. Limits lateral graph traversal and blocks circular overrides.
Tsukuri (Fitting)	Claim certification check. Validates marked payloads ([AXIOM], [SCHEMA], [BLUEPRINT], [LEAN]) against certified clerks.
Kake (Execution)	Action dispatch. Evaluates persist operations, quorum thresholds, and restores.
Ukemi (Receiving)	Thermodynamic stack health evaluation (<i>κ</i> , <i>H</i>). Freezes epoch and seals upon threshold breach.
Zanshin (Follow-through)	Telemetry recording, cryptographic digest chaining, and epoch advancement.

2. Locked Constants & System Parameters

Parameter	Locked Value
Active Lanes (<i>LANES</i>)	62
Quorum Threshold (<i>QUORUM</i>)	32
Default Hop Cap	8
Ukemi Kappa Threshold (κ)	0.85
Ukemi Energy Threshold (<i>H</i>)	0.90

Allowed Message Types

ingest, query, sibling, tool-result, persist, restore

Telemetry Caches

axiomatic, structural, dynamic, episodic

3. Security Rules & Invariants

- **Minuteman Role Isolation (RT-10):** Minuteman is strictly a structural role triggered upon policy violation, never a persistent agent ID in the agent registry.
- **Fail-Closed Persistence (RT-7):** Plaintext persistence is strictly prohibited. Episodic caches require a sealed cryptographic envelope (``bsai_master.key``).
- **No Marked Mints (RT-8):** Tool-result messages are prohibited from minting marked claims and cannot originate from the operator ID.
- **Lane Integrity (RT-3):** Writing into closed or quarantined lanes is blocked unconditionally.

4. Telemetry & Cryptographic Chain

Every transaction appends an immutable audit record to the episodic and dynamic caches. Each record includes SHA-256 digest chaining (*prev_digest*) to ensure tamper-evident logging across epochs.

```
{
  "ts": 1723849200.12345, "cache": "episodic", "agent_id": "oracle",
  "lane": 7, "C": 0.74000, "kappa": 0.21000, "H": 0.33000,
  "key_epoch": 1, "live_faces": 62, "hops": 1, "decision": "ALLOW",
  "resource": "query", "prev_digest": "00000000000000000000000000000000",
  "digest": "a3f8bc91e2047d12... [truncated 32 chars]"
}
```

5. Python Implementation Reference

The core specification logic is fully self-contained within the reference module `BastionShield`. It provides robust error handling, HKDF-derived envelope encryption via Python's ``cryptography`` library, and deterministic simulation execution via its built-in ``demo()`` harness.